

Exploratory Data Analysis (2)

July 9th, 2026

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Task

Understand structure

Summary statistics

Univariate distributions

Categorical analysis

Bivariate relationships

Correlation analysis

Group-wise summaries/Multivariate Visualization

Dimensionality reduction

Clustering and segmentation

Clustering visualization

Hypothesis generation

Tools & Techniques

~~.info(), .shape, .columns, .dtypes, .head()~~

~~.describe(), mean, median, std, IQR, min/max~~

~~Histograms, boxplots, density plots~~

~~Value counts, bar plots, pie charts~~

Scatter plots, grouped boxplots, violin plots, faceting

~~.corr(), heatmaps, pairplots~~

groupby, aggregation by category

PCA, t-SNE, UMAP (for visualization and structure discovery)

KMeans, DBSCAN, hierarchical clustering, Gaussian Mixture Models (GMM)

PCA/t-SNE scatter plots, cluster heatmaps, dendrograms, parallel coordinates

Observations and questions based on patterns or subgroup differences

Task 5: Bivariate relationship

- A **bivariate relationship** examines how **two variables** are related — whether one tends to change when the other does, or whether group membership affects outcomes.

Type 1

Type 2

Example

Questions to Ask

Numeric

Numeric

Age vs. Fare

Is there correlation? Linear or not?

Categorical

Numeric

Class vs. Fare

Do different classes pay different fares?

Categorical

Categorical

Sex vs. Survived

Are survival rates different by gender?

Pair Type

Recommended Plot(s)

Numeric vs. Numeric

Scatter plot, hexbin, jointplot

Categorical vs. Numeric

Boxplot, violin plot, strip plot

Categorical vs. Categorical

Grouped bar chart, heatmap, mosaic plot

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- Scatter Plot: Shows the **relationship** between two numeric variables



Pattern Seen

Upward trend

Flat/no pattern

Clusters

Outliers

Interpretation

Older passengers tend to pay higher fares

No clear relationship between age and fare

Possible subgroups (e.g., children, seniors)

A few paid unusually high fares (e.g., wealthy passengers)

- Joint Plot: Combines a **scatter plot** with **histograms or KDEs** (kernel density estimates) for both variables.
- LM Plot (Linear Model Plot): Plots a **scatter plot** with a **fitted regression line**.
- Hexbin: Divides the 2D space into **hexagonal bins** and **counts** how many points fall into each.



Feature

Meaning

Darker hexagons

More data points in that age-income range

Linear alignment

Suggests a trend or correlation

Scattered brightness

Indicates noise or spread

Empty regions

No observations in those ranges

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- Box Plot (Categorical vs. Numeric): Summarizes the distribution of a **numeric variable** grouped by a **categorical variable**.
- Violin Plot (Categorical vs. Numeric)
- Strip Plot (Categorical vs. Numeric): Plots **each individual observation** as a **dot** along a categorical axis; Helps you **see all values**, detect **clusters**, and spot **overlaps**.
- Swarm Plot (Categorical vs. Numeric): **neatly arranged strip plot, avoids overlap**
- Grouped Bar & Stacked Bar (Categorical vs. Categorical): Visualizes the **frequency or proportion** of combinations of two categorical variables.
- Heatmap of Crosstab (Categorical vs. Categorical): Summarizes **counts or proportions** of two categorical variables in a grid; Each cell shows how **often a combination occurs**; Heatmap coloring makes **differences easy to spot visually**.

More examples

- In practice, one does not have to use ALL methods
- Often one tries a few and pick the ones making more senses
- <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset/data>



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Task 6: Correlation analysis

- **Correlation analysis** measures the **strength and direction** of a linear relationship between pairs of **numeric variables**.
- Pearson Correlation Coefficient

Correlation Value

> 0.8 or < -0.8

$0.6 - 0.8$ or $-0.6 - -0.8$

$0.4 - 0.6$ or $-0.4 - -0.6$

$0.2 - 0.4$ or $-0.2 - -0.4$

≈ 0

Interpretation

Very strong

Strong

Moderate

Weak

Very weak or no linear relationship

Example

- Not much linear relationship
- A lot of linear relationship



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Task 7: multivariate visualization

Go beyond pairwise comparisons to **simultaneously explore multiple variables** (numeric and/or categorical), revealing:

- **Interactions between features**
- **Group patterns** across multiple dimensions
- Hidden **subgroup structures** that wouldn't show up in bivariate plots

Method

Type of Variables

Highlights

Enhanced Scatterplot

numeric + categorical

Quick overview of multiple variable influences

Pairplot with Hue

numeric + categorical

Class-wise exploration across many variables

FacetGrid / Catplot

numeric × categorical

Plots conditioned by group

Parallel Coordinates

numeric + class label

Feature profile comparison

3D Scatterplot (optional)

numeric × 3

Visualize tri-variable structure (interactively)

- Enhanced Scatterplot: A 2D scatterplot enriched with **hue**, **size**, and **style** encodings; Encode up to **4 variables** (x, y, color, size/shape); Ideal for continuous + categorical variable combinations
- Pairplot (with Hue): All pairwise scatterplots with **grouping information** overlaid; hue to highlight **subgroup differences**
- FacetGrid & Catplot: Grid of plots split by one or two **categorical variables**
- Parallel Coordinates: Each observation is a line plotted across multiple numeric axes.
- 3D Scatterplot

Example: Iris dataset

- **150 observations** of iris flowers
- **3 species** of iris:
 - *setosa*
 - *versicolor*
 - *virginica*
- **4 numeric features** measured in centimeters:
 - sepal length (cm)
 - sepal width (cm)
 - petal length (cm)
 - petal width (cm)



Iris Versicolor



Iris Setosa



Iris Virginica



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Task 8: Dimensionality Reduction & Structure Discovery

- **Dimensionality reduction** is the process of **transforming high-dimensional data into a lower-dimensional representation**, typically for downstream tasks
- To project high-dimensional data into **lower-dimensional space (2D or 3D)** in order to:
- **Visualize structure** that's hard to see in raw features
- **Identify clusters, patterns, or anomalies**
- **Prepare data** for further modeling (e.g. classification, clustering)

Type

Examples

Preserves...

Linear

PCA, MDS

Global structure
(variance/distance)

Manifold-based

Isomap, Laplacian Eigenmap,
*Diffusion Maps, *LLE

Geometry or topology of curved
spaces

Probabilistic

t-SNE, *UMAP

Local neighborhood structure

Neural

Autoencoders

Learned compressed
representation

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- PCA, MDS: looks for **orthogonal directions (axes)** in the data where the **variance is largest**
- Isomap, Laplacian Eigenmap, Diffusion Maps, Locally Linear Embedding: All four are **manifold learning** techniques that aim to **unfold or unroll** a nonlinear manifold embedded in high-dimensional space.
- t-SNE, U-map: Minimizes **Kullback-Leibler (KL) divergence** between high-D and low-D similarity distributions.
- Autoencoders: An **autoencoder** is a type of artificial neural network used to learn **efficient representations** (i.e., encodings) of data, typically for the purpose of **dimensionality reduction** or **denoising**.

Example

Dataset one: https://scikit-learn.org/1.5/auto_examples/datasets/plot_digits_last_image.html

Dataset two: https://rasbt.github.io/mlxtend/user_guide/data/wine_data/



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